Beginner Guide to AKS Storage Disks and Files

Kubernetes pods are temporary by default, meaning any saved data vanishes as soon as a pod restarts. Understanding Azure Disks and Azure Files in AKS ensures your application data stays safe and accessible no matter what happens to your pods.

What you'll learn:

1. Requesting Storage with PersistentVolumeClaims
2. Sharing Storage Across Pods with Azure Files
3. Mounting Storage Inside Container Folders
4. Troubleshooting Unmounted Storage Issues

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Requesting Storage with PersistentVolumeClaims

A PersistentVolumeClaim (PVC) acts like an order form for storage in Kubernetes. When you apply a PVC in AKS using the managed-csi storage class, Azure automatically creates a managed disk behind the scenes. This disk stays intact even if your pod dies or moves to another node.

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: azure-disk-pvc
spec:
  accessModes:
  - ReadWriteOnce
  storageClassName: managed-csi
  resources:
    requests:
      storage: 10Gi
```

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Sharing Storage Across Pods with Azure Files

Azure Disks can only attach to one pod at a time (ReadWriteOnce). If you have multiple web app replicas that need to share the same folder, use Azure Files with azurefile-csi. This creates a shared network file share accessible by all pods simultaneously (ReadWriteMany).

```
$apiVersion: v1
 kind: PersistentVolumeClaim
 metadata:
   name: azure-file-pvc
 spec:
   accessModes:
   - ReadWriteMany
   storageClassName: azurefile-csi
   resources:
     requests:
       storage: 5Gi
```

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Mounting Storage Inside Container Folders

Once your PVC is created, you must attach it to your app pod spec. You declare the PVC under the volumes section and link it to a container directory path under volumeMounts. Any file written to this folder will be stored safely in Azure.

```
apiVersion: v1
kind: Pod
metadata:
  name: my-app
spec:
  containers:
  - name: web
    image: nginx
    volumeMounts:
    - mountPath: "/mnt/azure"
      name: volume-data
  volumes:
  - name: volume-data
    persistentVolumeClaim:
      claimName: azure-disk-pvc
```

AKS Persistent Storage: Use Azure Disks & File Shares in Pods

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Troubleshooting Unmounted Storage Issues

When storage fails to attach, pods usually get stuck in a ContainerCreating or Pending status. Use kubectl commands to inspect PVC binding states and review event logs to quickly catch missing drivers, quota issues, or zone mismatched disks.

```
$kubectl get pvc,pv
 kubectl describe pvc azure-disk-pvc
 kubectl describe pod my-app
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

+ Use Azure Disks (managed-csi) for high-performance databases.
+ Use Azure Files (azurefile-csi) when multiple pods must read/write to the same folder.
+ Always set resource storage requests to avoid running out of disk space.
+ Delete unused PVCs to avoid paying for orphaned Azure Disks.
+ Check storage class defaults before creating custom PVC manifests.

Watch Out For

! Trying to attach a ReadWriteOnce Azure Disk to multiple pods across different nodes.
! Deleting a Pod and assuming its PVC will automatically delete itself.
! Forgetting to check node zone locations when attaching existing Azure Disks manually.

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